

Real-Time Bus Tracking and Passenger Occupancy Information System (RTBTIS)

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Abstract - The stochastic and unpredictability of urban traffic considerably hampers the dependability of public transportation, leading to fixed departure times ultimately failing during congestion and unfavourable weather conditions. A retrospective analysis of government-operated services reveals that average delays amount to 8 minutes, affecting the quality and reliability of transit operators' services and passenger satisfaction. This paper offers a comprehensive critique of the architectural growth of a Real-Time Bus Tracking and Passenger Occupancy Information System (RTBTIS), constructed to improve punctuality and safety, respectively.

The study delineates a layered technical evolution from economical smartphone telemetry to a resilient IPv6-based IoT system utilizing dedicated GSM channel for high-fidelity data acquisition of traffic. A combination of machine learning algorithms proves that the machine model outperforms traditional statistical models, providing state of art ETA accuracy where Support Vector Regression (SVR) trained on dynamic traffic and weather features delivers root mean square error as low as 20.8 seconds. This paper simultaneously evaluates the performance of deep computer-vision and AI-powered passenger monitoring detection of 99% accuracy and identifies the development of a transparent, fast, and consumer centric ecosystem as the primary deliverable of the technological advancement, followed by statistical query systems that provide key insights to transit authorities for utilizing optimal fleet sizes and valuable resources.

I. INTRODUCTION

Public transportation systems are crucial for urban sustainability. They help reduce congestion and lower emissions by moving large populations efficiently. Reliable transit is essential for modern Intelligent Transportation Systems (ITS). These systems aim to provide real-time information and service predictability similar to private companies. However, despite improvements in some areas, many urban bus networks, especially in developing cities, still struggle with unreliable arrival times. Factors like changing traffic, vehicle issues, and environmental conditions contribute to this.

Passengers often become frustrated and switch to private transport options. Traditional schedule-based systems usually do not handle these real-world uncertainties well, which leads to longer wait times and decreased public trust. To tackle these challenges, this research introduces an integrated Real-Time Bus Tracking and Passenger Occupancy Information System (RTBTIS). It aims to improve punctuality, reliability, and safety in bus networks. RTBTIS uses real-time data, smart tracking, and predictive analytics to give users live bus locations, updated Estimated Time of Arrival (ETA), and precise passenger occupancy data. ETA forecasting is improved with machine learning methods like Fully Connected Neural Networks and Support Vector Regression. These methods consider important factors such as traffic, vehicle data, time of day, and stop-level differences to reduce prediction errors.

Additionally, the system goes beyond just tracking vehicles. It enhances commuter safety and operational management by using Automatic Passenger Counting (APC) through Computer Vision and Artificial Intelligence (CV-AI) techniques. Modern camera-based and machine learning object detection models have shown accuracy rates of up to 99%. This allows transit authorities to manage resources and crowd control more effectively than older infrared or

Wi-Fi systems. The deployment occurs in two phases to ensure cost-effectiveness and scalability. The first phase uses mobile devices for tracking locations. The second phase adds specialized GPS/GSM modules and IoT sensors to support advanced machine learning models and increase system reliability. This research reviews current methods in real-time tracking, ETA prediction, and occupancy monitoring, providing a thorough and practical framework for developing RTBTIS to enhance urban mobility and service transparency.

II. LITERATURE SURVEY

Recent studies in 2025 focus on reducing commuting uncertainty in institutional environments through modular Internet of Things (IoT) architectures. Tahir et al. [1] initiated the focus on localized efficiency with the My Bus Tracker (MBT) system using GPS and Firebase. Complementing data reliability, Barabino et al. [2] proposed a framework for Automatic Passenger Counting (APC) in Open Mass Transit (OMT) utilizing video and passive infrared sensors. Karthik et al. [3] introduced "University Transport Connect," employing Flask-SocketIO for synchronization and machine learning for arrival predictions. Saravanan et al. [4] expanded transit intelligence by integrating IoT sensors with Support Vector Machine (SVM) algorithms to predict accident-prone zones. Prashanth et al. [5] developed a Kotlin and Android Jetpack framework facilitating GPS device communication, while Abdelhafed [6] utilized Waterfall methodology to create a modular tracking prototype enabling real-time congestion reporting.

Bijagare et al. [7] applied Kalman filtering and backpropagation to improve location precision and predict vehicle velocity. Yaazhini et al. [8] enhanced robustness by integrating weather and traffic into predictive models. Hardware efficiency was addressed by Agrahari and Azam [9] using Maduino Zero 4G LTE and AWS IoT, while Radovan et al. [10] identified YOLOv8 as the standard for passenger counting, emphasizing GDPR compliance. Akhade et al. [11] incorporated Kalman filtering and Firebase Authentication within the "Bus Nav" framework. Rashvand et al. [12] applied a Fully Connected Neural Network (FCNN) to NYC data, reducing arrival errors to under 40 seconds. Vision-based counting was refined by Prastyo et al. [13], utilizing YOLOv8 to demonstrate that front-view perspectives yield higher confidence (0.82) than rear-view placements (0.76).

Rozzaq et al. [14] used Raspberry Pi 4 and AI sensors for school bus density, noting crowding challenges. Talole et al. [15] implemented a centralized tracking system for the MSRTC. Ravindranath et al. [16] demonstrated continuously fine-tuning ML models using real-time weather and historical performance. Almeida et al. [17] characterized congestion using k-means clustering on OpenStreetMap data. Elijah et al. [18] proposed Bluetooth Low Energy (BLE) beacons detectable by Raspberry Pi Zero nodes at 20 meters as a low-cost alternative. Ahmed et al. [19] replaced MySQL with Firebase for real-time responsiveness, and Ashokan et al. [20] integrated the Google Maps Distance Matrix API and fee payment features.

Garigipati et al. [21] introduced "EASY BUS" with virtual passes and 30-second updates. Verma et al. [22] leveraged mobile sensors to detect aggressive driving, while Lohar et al. [23] designed fleet management with fire detection sensors. Somkuwar et al. [24] reviewed integrated technologies, highlighting the benefits of combining GPS with RFID-based ticketing. Shanathi et al. [25] achieved a Root Mean Square Error (RMSE) of 27 seconds using

Support Vector Regression (SVR) and a "jam factor." Ashwini et al. [26] proposed a hybrid XGBoost model distinguishing stochastic intersection delays. Abid et al. [27] designed a prototype utilizing Arduino Uno and SIM800L modules for secure HTTPS transmission. Noor et al. [28] compared ANN and SVM algorithms for travel prediction and environmental impact. Abdi and Amrit [29] classified prediction methods in a systematic review. Jain and Mishra [30] implemented an HF-RFID and NodeMCU system achieving a 0.985 detection probability and 161ms response time. Nitti et al. [31] introduced iABACUS using Wi-Fi probes for anonymous tracking with 94% accuracy. Prasanth et al. [32] and Thanuan et al. [33] utilized internal smartphone GPS for transit monitoring. Muthuselvi et al. [34] employed infrared sensors and the Haversine formula for 87.19% accuracy, while Korkmaz et al. [35] developed an encrypted application for school bus notifications.

Alamgir et al. [36] utilized Firebase for millisecond-level synchronization. Ameen et al. [37] analyzed cost-minimization strategies regarding SMS frequency, while Salam et al. [38] proposed a basic GPS/GPRS web interface. Weng et al. [39] achieved 91.4% speed estimation precision by matching GPS data to GIS arc sections. Kamel [40] implemented a microcontroller architecture using 2.5G GPRS. Mukhtar [41] and Ramani et al. [42] integrated remote vehicle control and alcohol sensors for safety. Mahalingam and Mahalakshmi [43] leveraged cloud computing and Google Maps APIs. Ganesh et al. [44] prioritized passenger information with voice tagging, while Chheda et al. [45] used RF transceivers for standalone LED displays. Medagama et al. [46] utilized open-source GIS tools for fleet management. Finally, Predic et al. [47] introduced motion prediction algorithms, and Brakatsoulas et al. [48] established foundational map-matching strategies based on Fréchet distance.

Modern vehicle tracking systems have evolved from basic satellite positioning into sophisticated, cloud-integrated ecosystems. By harmonising high-precision GPS sensors with low-latency communication protocols like Firebase and advanced predictive analytics, these systems now provide comprehensive mobility solutions that enhance operational efficiency, passenger safety, and user satisfaction in urban environments.

III. METHODOLOGY

1. REAL-TIME VEHICLE TRACKING AND FLEET MANAGEMENT METHODS

Early work in real-time vehicle monitoring established core algorithms for mapping raw spatial data to road networks and modelling motion. Brakatsoulas [48] introduced incremental and global map-matching methods using the Fréchet distance to address GPS noise and sampling inconsistencies. By 2011, research shifted toward localized monitoring with integrated GPS–GPRS modules and standalone controllers, such as the LED-based visualization system for Mumbai bus routes proposed by Chheda [45]. Mahalingam [43] and Mahalakshmi [43] and Ganesh [44] extended these systems with early Android interfaces and cloud hosting for real-time tracking and voice-based arrival announcements. In parallel, security-oriented solutions emerged, with Ramani [42] and Mukhtar [41] introducing GSM-based mechanisms for remote vehicle immobilization during theft between 2015 and 2020, systems matured through efficient communication techniques and low-cost embedded designs to reduce network overhead. Kamel [40] applied delta and run-length coding to compress location data, lowering GPRS usage and cost. Salam [38] and Medagama [46] further reduced hardware dependence by leveraging GSM infrastructure and open-source GIS tools such as MapServer for real-time fleet visualization. University-focused platforms were advanced by Alamgir [36], who replaced GSM hardware with Firebase-based synchronization, achieving millisecond-level updates. This trend continued with Korkmaz [35] and Muthuselvi [34], who integrated RFID and infrared sensors for passenger counting and route management within IoT frameworks. Smartphone-centric approaches by Thanuan [33] and Prasanth [32] utilized onboard GPS for cost-effective tracking, while Jain [30] and Mishra [30] showed that HF-RFID systems could outperform satellite tracking in update latency (161 ms). From 2022 onward, research emphasized high-precision positioning, traffic analysis, and administrative automation using advanced IoT nodes. Abid [27] and Ahmed [19] proposed NodeMCU-based architectures with GPRS connectivity for unreliable network environments

Table 1: Literature Survey on Real-Time Vehicle Tracking and Fleet Management Systems (2023–2025)

Sl. No	Authors	Year	GPS	GSM	IoT	Cloud	Web App	ML	ETA Support	Data Analytics	Technology Used
1	M. S. M. Tahir, K. W. Yi, S. Mohtar, R. R. Ikram	2025	✓	–	✓	✓	✓	–	✓	✓	GPS buses, Firebase cloud, Flutter app
2	B. Barabino, L. Raccagni, R. Ventura	2025	–	–	–	–	–	–	–	–	APC accuracy using IR & video sensors
3	A. Radovan, L. Mršić, G. Đambić, B. Mihaljević	2024	✓	–	✓	✓	✓	✓	–	✓	YOLO-based APC with IoT & CCTV
4	N. Rashvand, S. S. Hosseini, M. Azarbayjani, H. Tabkhi	2024	✓	–	–	✓	–	✓	✓	–	FCNN-based ETA prediction
5	P. Agrahari, I. Azam	2024	✓	✓	✓	✓	–	–	–	–	GPS + 4G LTE + AWS IoT
6	A. Almeida, S. Brás, S. Sargento, I. Oliveira	2023	✓	–	✓	✓	✓	–	–	✓	Smart city buses with clustering
7	O. Elijah, S. L. Keoh, S. K. Rahim, C. K. Seow	2023	–	–	✓	✓	–	–	✓	✓	BLE beacon-based bus tracking
8	M. K. Ahmed, M. A. Islam, M. A. Iqbal,	2023	✓	✓	✓	✓	✓	–	–	–	NodeMCU GPS with Firebase
9	M. Ashokan, N. Niranjana, M. KY, R. Prasanna	2023	✓	✓	✓	✓	–	–	✓	–	College bus GPS tracking

To address congestion, Almeida [17] applied k-means clustering to bus probe data combined with OpenStreetMap, transforming noisy GPS readings into actionable traffic insights. System robustness was enhanced by Ashokan [20] and Garigipati [21] through Google Maps Distance Matrix integration and virtual bus pass verification. Lohar [23] and Akhade [11] improved coordinate stability using Kalman filtering and linear regression, while Agrahari [9] and Karthik [3] introduced 4G LTE Arduino Zero boards and Flask-SocketIO for low-latency client-server communication and predictive analytics. Recent studies focus on proactive safety and comprehensive fleet analytics through multi-source data fusion. Rozzaq [14] and Ravindranath [16] employed computer vision and machine learning to monitor passenger density and update arrival forecasts using live weather data. Saravanan [4] developed SVM-based models to identify accident-prone zones from historical traffic patterns. To improve operational reliability, Talole [15] and Tahir [1] implemented role-based mobile interfaces with authentication and route-deviation alerts. Accuracy was further enhanced by Bijagare [7] and Yaazhini [8] using ensemble ML methods, while Abdelhafed [6] and Elijah [18] demonstrated that combining redundant sensing, such as Bluetooth Low Energy beacons, improves resilience against signal loss and spoofing. Overall, modern vehicle tracking systems now integrate precise positioning, low-latency communication, and cloud-based analytics to deliver comprehensive and reliable mobility solutions.

2.ETA PREDICTION AND PASSENGER ANALYTICS METHODS

Early research established the use of Automatic Vehicle Location (AVL) data and historical motion profiles to address the instability of arrival time estimates caused by urban traffic stochasticity. Predic [47] demonstrated that statistical analysis of repeating historical patterns in vehicle advancement could provide more stable long-term predictions than those based solely on current motion parameters like instantaneous speed. Similarly, Ganesh [44] developed a prototype that utilized link travel times derived from periodic GPRS updates, implementing a simple algorithm that assigned lesser weight to older observations to better reflect current traffic conditions. These foundational methods highlighted the necessity of filtering and classifying data based on representative time frames, such as rush hours and weekends, to improve reliability.

By the mid-2010s, models began incorporating geographical information system (GIS) maps and sophisticated distance calculations to enhance precision. Weng [39] introduced a real-time

travel speed estimation model for 20,000 vehicles in Beijing, utilizing discrete GPS spot data and specialized processing procedures to match bus locations accurately onto arc sections. This approach achieved a precision of 91.4% by modifying stop positions to account for systematic errors in original GPS data. Muthuselv [34] further refined these calculations for urban environments by comparing Euclidean and Haversine formulas for distance estimation, finding that Haversine-based arrival time predictions achieved a superior average accuracy of 87.19%. Ahmed [19] extended these architectures by leveraging Firebase for real-time database synchronization, providing arrival estimates within milliseconds.

Recent literature has shifted toward machine learning (ML) to capture the non-linear relationships inherent in complex transit networks. Noor [28] and Shanthi [25] conducted comparative studies between Artificial Neural Networks (ANN) and Support Vector Regression (SVR), observing that ANN models consistently outperformed SVR in travel time predictive accuracy. Shanthi [25] specifically enhanced prediction precision by integrating environmental dynamics, such as weather conditions and a "jam factor" representing traffic density, into the SVR framework, achieving a Root Mean Square Error (RMSE) of 20 seconds. Addressing spatial variability, Ashwini [26] developed individual XGBoost models for different road patterns, demonstrating that sections without intersections are significantly more predictable than those influenced by stochastic delays at signalled intersections.

The most recent advancements prioritise high-dimensional deep learning (DL) and ensemble methods to ensure system scalability across large metropolitan networks. Rashvand [12] developed a unified Fully Connected Neural Network (FCNN) framework using New York City bus data, proving that FCNNs maintain error margins under 40 seconds across 232 lines, whereas traditional SVR models become untrainable as the number of transit lines increases. Concurrently, Bijagare [7] and Karthik [3] have focused on refining coordinate stability and responsiveness through Kalman filtering, linear regression smoothing, and Flask-SocketIO for seamless client-server communication. These modern systems synthesize multi-source data including live meteorological logs, traffic sensors, and historical performance to transition from static schedules toward intelligent, real-time urban mobility solutions. Overall, modern arrival prediction systems provide a comprehensive mobility solution by harmonising high-precision positioning sensors with low-latency communication protocols and deep learning predictive analytics.

Table 2: Literature Survey on ETA Prediction & Data-Driven Bus Tracking Systems (2022–2025)

Sl. No	Authors	Year	GPS / AVL	Model / Algorithm	Dataset Used	ETA Output	Technology Used	Accuracy (%)
1	Vivek Bijagare, S. Patil, M. Rabade	2025	✓	Base algorithm based on historical journey time	BLE Proximity Sensing (two public bus services in southern Malaysian cities)	✓	GPS + Kalman + Linear Regression	–
2	Karthik C, P. Kumar N	2025	✓	Fully Connected Neural Network (FCNN)	New York City (NYC) Bus System Dataset	✓	Linear Regression + GPS	RMSE of 35.74 seconds
3	C. Saravanan, B. Ayyappa	2025	✓	–	–	✓	IoT + SVM + Live GPS	–
4	N. Rashvand, S. S. Hosseini	2024	✓	Support Vector Regression (SVR)	Real-time tracking data from Kongu Engineering College bus route	✓	FCNN on NYC Bus Dataset	RMSE of 20 seconds
5	K. Akhade, K. Dadge, R. Nikam	2024	✓	Predictive Analytics (Flask/WebSockets)	University bus tracking data	✓	GPS + Kalman + Speed History	–
6	O. Elijah, S. L. Keoh, S. K. Rahim	2023	✓	Support Vector Machine (SVM)	Bluetooth proximity beacons for public bus tracking	✓	BLE Beacons + Historical ETA	90% (for accident zone risk classification)
7	N. Shanthi, S. V. E. Sathishkumar	2022	✓	Neural Network	151 bus routes in Boston	✓	SVR + GPS + Weather	80 seconds deviation
8	B. P. Ashwini, R. Sumathi, H. S. Sudhira	2022	✓	Data Analytics	Historical accident data (GPS coordinates)	✓	XGBoost + Historical GPS	–

3. PASSENGER OCCUPANCY AND CROWDING ESTIMATION METHODS

The evolution of Automatic Passenger Counting (APC) has transitioned from basic physical sensing to complex, non-intrusive data fusion frameworks integrated with the Internet of Things (IoT). Early implementations relied on cost-effective physical sensors to establish basic occupancy monitoring. Muthuselvi [34] (2019) developed an intelligent system using infrared (IR) sensors placed at entry and exit doors to increment and decrement passenger totals via an Arduino controller, enabling real-time overcrowding alerts. While foundational, these early sensor-based methods including pressure mats and IR beams remained susceptible to inaccuracies during peak boarding hours when high passenger density caused object overlap or signal blocking.

The mid-stage of APC development saw a shift toward non-intrusive wireless tracking to mitigate hardware maintenance costs. Nitti [31] (2020) introduced the iABACUS framework, which utilizes Wi-Fi probe requests to infer passenger numbers. A significant technical advancement of this system was its de-randomization algorithm, which successfully addressed MAC address randomization in modern mobile operating systems to ensure accurate tracking from boarding to alighting. Despite achieving nearly 94% accuracy in dynamic tests, this approach faced challenges regarding signal attenuation caused by human bodies in crowded environments. Recent literature emphasizes the dominance of vision-based systems

powered by deep learning for robust real-time density estimation. Radovan [10] (2024) and Prastyo [13] (2024) highlighted the transition toward Convolutional Neural Networks (CNN) and YOLO architectures, which excel at identifying individuals in complex, high-density settings. Specifically, the YOLOv8 model demonstrated superior efficiency, though its performance remained sensitive to camera perspective; front-view cameras achieved higher confidence scores (0.82) than rear-view placements (0.76) due to reduced object overlap. To further refine accuracy, Rozzaq [14] (2024) integrated Raspberry Pi 4 processing with webcam-based human detection, achieving 97% accuracy in school bus monitoring, though they noted that crowded bus entrances still necessitate improvements such as face identification.

The most recent advancements prioritise integrated, multi-modal frameworks and privacy-aware sensing. Barabino [2] (2025) proposed a framework for Open Mass Transit (OMT) that combines video acquisition with passive infrared sensors, finding that while system reliability is high, accuracy is significantly influenced by station architecture and independent access points. Contemporary systems are increasingly designed around GDPR compliance, employing techniques such as face blurring, data minimization, and edge processing to safeguard passenger privacy while maintaining operational efficiency. Overall, modern APC systems integrate multi-sensor or vision-based approaches with IoT platforms to enable reliable passenger density monitoring and service optimization.

Table 3: Literature Survey on Passenger Occupancy and Crowding Estimation Techniques (2022–2025)

Sl. No	Authors	Year	Sensor-Based APC	Model / Algorithm	Dataset Used	Density / Crowding Estimation	Technology Used	Accuracy (%)
1	B. Barabino, L. Raccagni, R. Ventura	2025	✓	CNN	CCTV passenger data	✓	IR sensors + video-based APC	92–99%
2	A. Radovan, L. Mršić, G. Đambić	2024	✓	YOLO v8 / Modified ByteTrack	CrowdHuman benchmark dataset	✓	CCTV + YOLO/CNN + multi-sensors	–
3	R. G. Ravindranath, V. Korubilli, N. Ratnala	2024	✓	Video-based automatic people counting	On-bus and Off-bus deployment data	✓	Sensors + ML-based occupancy	–
4	R. I. R. Nugroho, G. H. Panjaitan, R. H. Taloputri	2024	–	YOLO v8	COCO (Common Objects in Context)	✓	Webcam + AI on Raspberry Pi	–
5	A. D. Prastyo, S. A. Minhulina, S. Agung	2024	–	ANN and LSTM	Xingtai bus company dataset	✓	CCTV / smartphone + YOLOv8	88–89%
6	O. Elijah, S. L. Keoh, S. K. Rahim	2023	✓	BLE proximity detection	Public bus fleet data (Johor, Malaysia)	✓	BLE beacons + ANN (MLP)	–
7	M. Kassim, A. S. Salleh	2022	✓	YOLO v5 / DeepSORT	Large-scale bus passengers' dataset	–	IR sensors + GPS-RFID	96.6%

IV. RESULTS

Recent research demonstrates a clear shift from basic GPS-based vehicle tracking toward integrated, analytics-driven intelligent transportation systems. Studies by Ahmed et al. [19], Ashokan et al. [20], and Prashanth et al. [5] show that NodeMCU and smartphone-based tracking combined with Firebase cloud platforms enables low-cost, cross-platform monitoring for institutional transit networks. While Ahmed et al. [19] prioritize accessibility across web and mobile interfaces, Ashokan et al. [20] improve arrival-time estimation by incorporating road-network-aware services such as the Google Maps Distance Matrix API. Extending beyond location tracking, Garigipati et al. [21] integrate virtual bus pass automation, indicating a broader trend toward administrative and user-service integration. In contrast, Elijah et al. [18] propose a proximity-based sensing paradigm using Bluetooth Low Energy (BLE) beacons, achieving improved positioning reliability in dense urban environments where GPS performance degrades. Complementing these system-level advances, Almeida et al. [17] apply clustering techniques to aggregated tracking data, demonstrating that fleet data can support higher-level congestion analysis and urban planning.

Despite these advances, most 2023 frameworks remain constrained by their reliance on stable Wi-Fi or cellular connectivity, which continues to limit seamless real-time synchronization. Earlier and foundational research emphasizes the role of advanced modelling techniques and alternative architectures in improving arrival time prediction and system responsiveness. Shanthi et al. [25] and Ashwini et al. [26] demonstrate that non-linear machine learning models such as Support Vector Regression and XGBoost outperform static schedule-based and linear statistical approaches under dynamic traffic conditions. Hardware-oriented studies by Abid et al. [27] advocate GPRS-based transmission over SMS to reduce latency and operational costs. A fundamentally different architecture is introduced by Jain and Mishra [30], who replace GPS-based tracking with HF-RFID tagging at bus stops to achieve sub-second update latency, outperforming smartphone-based polling systems discussed by Prasanth et al. [32] and Thanuan et al. [33]. In the context of passenger analytics, Nitti et al. [31] propose Wi-Fi-based anonymous sensing to address privacy challenges inherent in vision-based Automatic Passenger Counting systems evaluated by Barabino et al. [2]. Finally, the survey by Abdi and Amrit [29] concludes that although machine learning dominates modern arrival-

time prediction research, the lack of real-time multi-source data fusion particularly for weather, driving behaviour, and passenger flow remains a critical unresolved challenge.

V. DISCUSSION

The evolution of vehicle tracking demonstrates a clear shift from passive systems, which required physical access to internal storage, and manual surveys to real-time, IoT-integrated architectures leveraging cloud-based platforms such as Firebase and AWS. While foundational GPS-GSM models established basic location acquisition, newer frameworks prioritising predictive intelligence address the inherent inaccuracies of static schedules and expensive, high-latency SMS reporting. Evidence from recent studies indicates that non-linear architectures consistently outperform classical statistical models; for instance, Rashvand [12] demonstrated that Fully Connected Neural Networks (FCNN) maintain high accuracy and scalability, whereas traditional Support Vector Regression (SVR) becomes untrainable as the number of transit lines increases. Similarly, Noor [28] and Shanthi [25] found that Artificial Neural Networks (ANN) and regression models that integrate environmental variables—including weather conditions and traffic jam factors—provide significantly more robust arrival time estimates than traditional linear methods.

Despite these technical advancements, system performance remains constrained by environmental and structural factors. While Elijah [18] observed that proximity-based BLE beacons offer a cost-effective and reliable alternative to GPS in dense urban corridors, satellite-based systems continue to face signal loss in buildings and RF-shadowed areas. In the domain of occupancy management, vision-based counting models such as YOLOv8 report high precision but are limited by perspective sensitivity and object overlap in crowded bus entrances, where density leads to systematic undercounting. Furthermore, Ahmed [19] and Bijagare [7] emphasise that the reliability of real-time synchronisation is fundamentally dependent on the stability of regional cellular network infrastructure and continuous power supply, which remain primary technical barriers to seamless deployment.

VI. FUTURE WORKS

Future research will focus on advancing sensing robustness, predictive precision, and system infrastructure to facilitate the transition from localized prototypes to city-wide intelligent transportation ecosystems.

- **Multimodal Sensor Fusion:** Integrating computer vision with Wi-Fi sniffing and infrared sensors to reduce counting inaccuracies caused by passenger occlusion in high-density environments.
- **Environmental and Contextual Integration:** Improving forecasting reliability by incorporating real-time meteorological data, traffic incidents, and delays at signalised intersections.
- **Advanced Predictive Architectures:** Adopting self-attention-based neural networks and ensemble models to better capture long-term temporal dependencies and enhance arrival time accuracy across diverse road geometries.
- **Energy and Hardware Optimization:** Utilizing sustainable power solutions such as solar harvesting and offloading computation to backend servers to minimize processing load on edge microcontrollers.
- **Resilient Security and Privacy:** Enhancing protection against Denial of Service (DoS) attacks and applying advanced anonymization techniques, including de-randomization filters, to ensure compliance with data protection regulations.

VII. CONCLUSION

Recent research indicates that non-linear deep learning architectures and multimodal fusion provide the most reliable transit monitoring outcomes. Rashvand et al. [12] found that Fully Connected Neural Networks (FCNN) maintain arrival time error margins under 40 seconds, vastly outperforming traditional Support Vector Regression (SVR) models which lack scalability across large bus networks. Similarly, Shanthi et al. [25] and Noor et al. [28] observed

that Artificial Neural Networks (ANN) consistently yield superior travel time predictive accuracy over linear models when factors such as weather and traffic "jam factors" are integrated. In occupancy management, Prastyo et al. [13] and Radovan et al. [10] reported that YOLOv8 achieves precision levels exceeding 95%, outperforming mid-range results by better capturing the stochastic nature of urban boarding flows.

Comparative outcomes reveal that dataset difficulty significantly affects reliability; Radovan et al. [10] noted that system accuracy often declines by 20% to 30% in real-world settings compared to laboratory conditions. Prastyo et al. [13] distinguished top-performing front-view camera perspectives from moderate rear-view placements, which suffer from object overlap in crowded situations. For positioning, Elijah et al. [18] demonstrated that proximity-based BLE beacons offer a robust alternative to traditional GPS in urban canyons, while Jain and Mishra [30] achieved high-speed updates of 161ms using RFID. Furthermore, Muthuselvi et al. [34] found that Haversine-based distance estimation achieves 87.19% accuracy, significantly surpassing results obtained via Euclidean formulas. Ashwini et al. [26] also identified that road sections without intersections provide higher predictability than those influenced by stochastic delay patterns.

The key takeaway is that the fusion of real-time environmental variables and low-latency IoT protocols consistently leads to enhanced system performance. However, limitations reported by Kamel [40] include conventional GPS signal loss in RF-shadowed areas, while Nitti et al. [31] and Barabino et al. [2] noted random errors at closely spaced stops and slight overestimations in open mass transit. Authors such as Radovan et al. [10] emphasise that addressing these constraints requires balancing high computational demands with strategic sensor placement. Ultimately, the synergy between advanced predictive models and resilient communication hardware defines the state-of-the-art in modern fleet management.

VII. REFERENCES

1. M. S. M. Tahir, K. W. Yi, and S. Mohtar, et al., "Smart Mobility on Campus: A Real-Time GPS Bus Tracking System With Cloud Integration," 2025.
2. B. Barabino, L. Raccagni, and R. Ventura, "Accuracy of Automatic Passenger Counting in Open Mass Transit Systems: Insights from Italy," 2025.
3. Karthik C., Praveen Kumar N., and S. S. Clarinda, "University Transport Connect: Real-Time University Bus Tracking System Using Flask Websocket And Machine Learning Predictive Analysis," 2025.
4. C. Saravanan, B. Ayyappa, and R. Bhuvaneswaran, et al., "Real-Time Bus Tracking and Management System with IOT, Chatbot Integration and Accident Zone Prediction," 2025.
5. Prashanth H. S., Adithya M., and Achyutha U. N., et al., "Real-Time Transportation Tracking: A GPS Enabled Mobile Application Framework," 2025.
6. Y. H. A. Abdelhafed, "Improving Transportation Management by Real-Time Bus Tracking using GPS," 2025.
7. V. Bijagare, S. Patil, and M. Rabade, et al., "Busnav A Real Time Bus-Tracking App," 2025.
8. M. A. Yaazhini, Ramnarayanan A., and Ram Kumar R., et al., "Smart Bus Tracking And Notification System Using Machine Learning For Enhanced Urban Mobility," 2025.
9. P. Aghari and I. Azam, "Real-Time Bus Tracking System," 2024.
10. A. Radovan, L. Mršić, and G. Dambić, et al., "A Review of Passenger Counting in Public Transport Concepts with Solution Proposal Based on Image Processing and Machine Learning," 2024.
11. K. Akhade, K. Dadge, and R. Nikam, et al., "Bus Nav A Real Time Bus Tracking Application," 2024.

12. N. Rashvand, S. S. Hosseini, and M. Azarbayjani, et al., "Real-Time Bus Arrival Prediction: A Deep Learning Approach for Enhanced Urban Mobility," 2024.
13. A. D. Prastyo, S. A. Minhalina, and S. Agung, et al., "Automatic Passenger Counting System on Public Buses Using CNN Model for Passenger Capacity Optimization," 2024.
14. R. I. R. Nugroho, J. G. H. Panjaitan, and R. H. Taloputri, et al., "Empowering the capabilities and Service Optimization for Location Tracking and Management of School Bus Passenger Density in Jakarta through the Integration of CV-AI," 2024.
15. V. Talole, P. Sonavane, and S. Saykar, et al., "Implementation of Real-Time Bus Tracking and Passenger Information System for MSRTC," 2024.
16. Ravindranath G., V. Korubilli, and N. Ratnala, et al., "Smart Bus Tracker: Gps-Driver Arrival Forecasting For Bus Routes," 2024.
17. A. Almeida, S. Brás, and S. Sargento, et al., "Exploring bus tracking data to characterize urban traffic congestion," 2023.
18. O. Elijah, S. L. Keoh, and S. K. b. A. Rahim, et al., "Transforming urban mobility with internet of things: public bus fleet tracking using proximity-based bluetooth beacons with Android and Web interfaces," 2023.
19. M. K. Ahmed, M. A. Islam, and M. A. Iqbal, et al., "A Cross-Platform Vehicle Tracking System for Pabna University of Science and Technology," 2023.
20. M. Ashokan, N. Niranjana, and Musafir K. Y., et al., "Campus Connect-A College Bus Tracking System," 2023.
21. N. Garigipati, C. S. Reddy, and C. S. Charitha, et al., "Easy Bus (Virtual Bus Pass and Bus Tracking System) Design and Implementation Using Machine Learning Algorithms," 2023.
22. K. K. Verma, B. M. Singh, and Mridula, "Vehicle tracking system using working sensors of a partially functional android mobile phone," 2023.
23. P. Lohar, S. Khillare, and T. Ghodke, et al., "Design and Implementation of Vehicle Tracking System with GPS," 2023.
24. P. Somkuwar, A. Amune, and S. Mehtre, et al., "Literature Review: Real-Time Bus Tracking, RFID-Based Bus Ticketing, and RFID for Door Access Control," 2023.
25. N. Shanthi, Sathishkumar V. E., and K. U. Babu, et al., "Analysis on the Bus Arrival Time Prediction Model for Human-Centric Services Using Data Mining Techniques," 2022.
26. B. P. Ashwini, R. Sumathi, and H. S. Sudhira, "A Dynamic Model for Bus Arrival Time Estimation based on Spatial Patterns using Machine Learning," 2022.
27. M. H. Abid, A. Islam, and A. D. Biswas, et al., "IoT-Based Vehicle Tracking System For Khulna University," 2022.
28. R. M. Noor, N. B. G. Rasyidi, and T. Nandy, et al., "Campus Shuttle Bus Route Optimization Using Machine Learning Predictive Analysis: A Case Study," 2021.
29. A. Abdi and C. Amrit, "A review of travel and arrival-time prediction methods on road networks: classification, challenges and opportunities," 2021.
30. A. Jain and A. Mishra, "Design of IoT based Real-Time Bus Tracking App using HF-RFID," 2021.
31. M. Nitti, F. Pinna, and L. Pintor, et al., "iABACUS: A Wi-Fi-Based Automatic Bus Passenger Counting System," 2020.
32. T. Prasanth, Shravya K., and Bharath M. S., et al., "College Bus Tracker Android Application," 2020.
33. S. Thanuan, A. Maurya, and A. Chahal, et al., "Vehicle Tracking System using GPS and Android," 2020.
34. M. Muthuselvi, M. A. Abi, and V. A. Arthi, et al., "An Intelligent Bus System Based on Internet of Things for Urban Environments," 2019.
35. I. Korkmaz, D. Dirik, and A. Camci, et al., "A Smart School Bus Tracking System," 2019.
36. M. S. Islam, M. Mynuddin, and M. A. Ullah, et al., "Chittagong University Teachers Bus Tracking System Using Smartphone Application," 2018.
37. F. N. Ameen, Z. S. Mohammed, and A. I. Siddiq, "Cost Minimization of GPS-GSM Based Vehicle Tracking System," 2018.
38. A. Salam, I. M. Chowdhury, and M. M. Sadeque, et al., "Save Time for Public Transport Users in a Developing Country," 2016.
39. J. Weng, C. Wang, and H. Huang, et al., "Real-time bus travel speed estimation model based on bus GPS data," 2016. ³⁹
40. M. B. M. Kamel, "Real-Time GPS/GPRS Based Vehicle Tracking System," 2015.
41. M. Mukhtar, "GPS based Advanced Vehicle Tracking and Vehicle Control System," 2015.
42. R. Ramani, S. Valarmathy, and N. SuthanthiraVanitha, et al., "Vehicle Tracking and Locking System Based on GSM and GPS," 2013.
43. T. Mahalingam and M. Mahalakshmi, "Android-A Cloud Computing For Vehicle Tracking System Using Gps," 2012.
44. Ganesh K., Thirvikraman M., and J. Kuri, et al., "Implementation Of A Real Time Passenger Information System," 2012.
45. G. Chheda, N. Gajra, and M. Chhaya, et al., "Real Time Bus Monitoring and Passenger Information System," 2011. ⁴⁵
46. M. Medagama, D. Gamage, and L. Wijesinghe, et al., "GIS/GPS/GPRS and Web based Framework for Vehicle Fleet Tracking," 2008.
47. B. Predic, D. Rancic, and D. Stojanovic, et al., "Automatic vehicle location in public bus transportation system," 2007. ⁴⁷
48. S. Brakatsoulas, D. Pfoser, and R. Salas, et al., "On Map-Matching Vehicle Tracking Data," 2005.